

DEVELOPMENT OF A NEW TYPE FAULT LOCATOR
USING THE ONE-TERMINAL VOLTAGE AND CURRENT DATA

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Abstract A new type fault locator has been developed, that calculates the reactance of a faulty line, with a micro-processor, using the one-terminal voltage and current data of the transmission line. Errors caused by various factors such as load flow, fault resistance, and the unsymmetrical arrangement of the transmission line, are automatically corrected. The equipment is now under field test, and during the nine months of the field test, the equipment has operated satisfactorily.

INTRODUCTION

Fault Locators, that measure the distance to a fault point on a transmission line, conventionally use the following methods. One is to detect the surge voltage caused by a fault at both sides of the transmission line and the time difference of the surge detected between the two terminals determines the fault point. Another method is, to set off pulse signals when a fault occurs and the pulse return time from the fault point determines the fault point. Recently, attempts to apply micro-processors to control and protection facilities of electric power systems have increased and in some fields, such facilities are in commercial use. The fault locator equipment is also expected to measure the distance more precisely and easily, by micro-processor, using digitized line voltage and current data.

In calculating the distance to a fault point, using voltage and current data, it is obviously advantageous to use both sides of the terminal data of the faulty transmission line. However, from the practical viewpoint, simple equipment is desirable, and, it is very important that the equipment should use only one terminal voltage and current data.

A reactance relay, used conventionally in protective relaying equipment, functions to calculate the distance to a fault point using the one-terminal voltage and current input. However, a reactance relay cannot get precise results for the following reasons:

- i) The results are influenced by the load flow and the fault resistance.
- ii) The results are also influenced by the current flowing through other phases of the transmission line by mutual impedance.

The digital fault locator (DFL), described in this paper, uses a micro-processor and reduces errors by an efficient use of software.

PRINCIPLE

Fundamental Equation

The method to calculate the distance to a fault point is based on the following equation, that expresses a fault point voltage V_F and current I_F using the one-terminal data.

$$\begin{aligned} V_F &= V_S \cosh \gamma x - Z_S I_S \sinh \gamma x \\ I_{SF}'' &= V_S' / Z_S \sinh \gamma x - I_S'' \cosh \gamma x \end{aligned} \quad \dots (1)$$

For (1), two approximations are adopted:

- i) $\tanh \gamma x \approx \gamma x$, because a transmission line is sufficiently short.
- ii) angles of a fault current I_F and a fault current from the S-terminal I_{SF}'' are equal.

Then, the distance x is obtained using the one-terminal data as in (2). Details are shown in Appendix I.

$$x = \frac{\text{Im} (V_S \cdot I_{SF}'')}{\text{Im} (Z_S I_S \cdot I_{SF}'')} \quad \dots (2)$$

where

- x : distance to the fault point
- V_S : voltage of S-terminal
- I_S : current of S-terminal
- V_F : voltage of fault point
- I_F : fault current
- V_S' : voltage difference between pre-fault and after-fault voltage.
- I_S'' : current difference between pre-fault and after-fault (fault component current)
- I_{SF}'' : fault current from S-terminal
- Z_S : surge impedance
- Z : transmission line impedance per unit length
- γ : propagation constant
- Im : imaginary component
- $*$: conjugate component

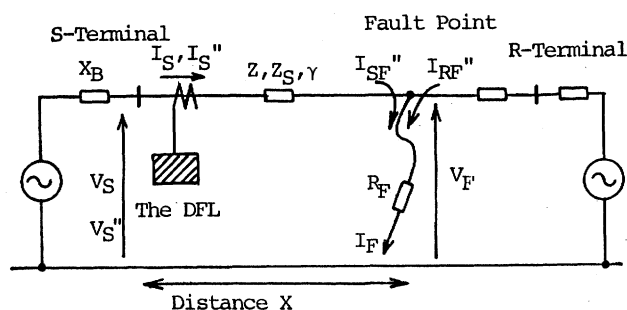


Fig. 1 Fault in a single phase circuit

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In equation (2), effect of the load flow is cancelled by using fault component current I_F^s , and the effect of fault resistance is reduced by eliminating fault resistance R_F (cf. appendix I).

The methods to reduce other error factors are also described.

Effect of Other Phase Currents

In applying equation (2) to fault locating, we must pay attention to induced voltage from other phases or other transmission lines on the same tower. Mutual impedance between each phase is determined by the arrangement of each phase line, and, it is nonsymmetrical. In the fault locator, which needs precise results, this nonsymmetry cannot be neglected. The fault locator, we developed, covers two parallel lines furnished to the same tower, and the effect of the other phase currents is compensated by the following equation, according to the kind of fault.

For No. 1 line R-phase to ground fault:

$$x = \frac{\text{Im}(V_{R1} \cdot I_{\alpha R1}^{**})}{\text{Im}(V_{LR1} \cdot I_{\alpha R1}^{**})} \quad \dots (3)$$

$$V_{LR1} = \sum_K (Z_{R1k} \cdot I_k)$$

$$k = R1, S1, T1, R2, S2, T2$$

Where $R1, S1, T1$ means each phase of No.1 line

$R2, S2, T2$ means each phase of No.2 line

V_{R1} : NO.1 line R phase voltage

$I_{\alpha R1}$: α component of No.1 line R-phase current

Z_{R1k} : Mutual impedance concerning No.1 line R phase

For No.1 line S phase to T phase fault

$$x = \frac{\text{Im}(V_{S1T1} \cdot I_{S1T1}^{**})}{\text{Im}(V_{LS1T1} \cdot I_{S1T1}^{**})}$$

$$V_{LS1T1} = (Z_{S1S1} - Z_{S1T1}) I_{S1} - (Z_{T1T1} - Z_{S1T1}) I_{T1} + (Z_{R1S1} - Z_{R1T1}) I_{R1} + \sum_K (Z_{S1k} - Z_{T1k}) I_k$$

$$k = R2, S2, T2 \quad \dots (4)$$

V_{S1T1} : No.1 line S-T phase Δ voltage

I_{S1T1} : No.1 line S-T phase Δ current

Two-phases-to-ground faults and three phase faults are included in (4).

Approximation Error

The following approximation is used in deriving the fundamental equation (2).

$$\tanh \gamma x \approx \gamma x \quad \dots (5)$$

The effect of this approximation is very small, and it can be neglected when the transmission line is less than 100km long. However, for a line more than 100km long, the following compensation is effective. Figure 2 shows an example of the approximation error and the compensation effect. Details are shown in Appendix II.

$$x_2 = x_1 - \beta^2 \cdot x_1^3 / 3 \quad \dots (6)$$

x_1 : calculation result by (2)

β : $I_m(\gamma)$, γ is propagation constant

x_2 : compensated result

Concerning the approximation that the angles of a fault current I_F and a fault current from the S-terminal I_{FS}'' are equal, the effect is negligible, because fault impedances are resistive.

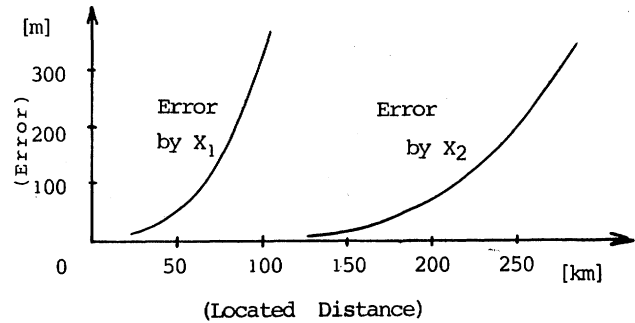


Fig. 2 An example of approximation error

Effect of Fault Current in an Overhead Ground Wire

Between each tower of the transmission line, an overhead ground wire is stretched that has the same voltage to the tower, and each tower is grounded through a tower-footing resistance.

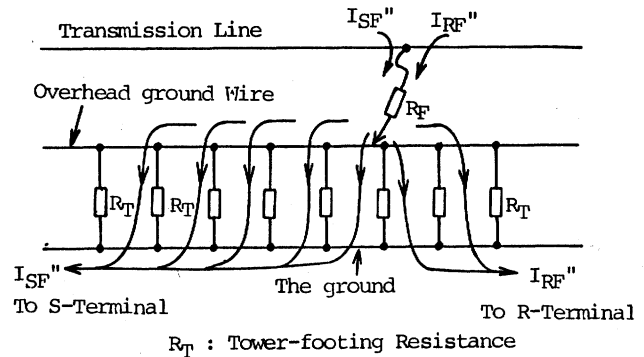


Fig. 3 Flow of a ground fault current

When a ground fault occurs, most of the fault current flows to the overhead ground wire, and the longer the distance from the fault point becomes, the more fault current flows to ground through tower-footing resistances. Zero phase impedance, used in the fault locator calculation, is decided with the condition that the ratio of the ground current and the overhead ground wire current is constant. Error derives from the fact that this ratio is not constant near the fault point. This error can be calculated by solving the equation shown in Appendix III. Figure 4 shows the calculated error for the Naka-Tokyo Line where the DFL is located, and this error is automatically corrected by the micro-processor.

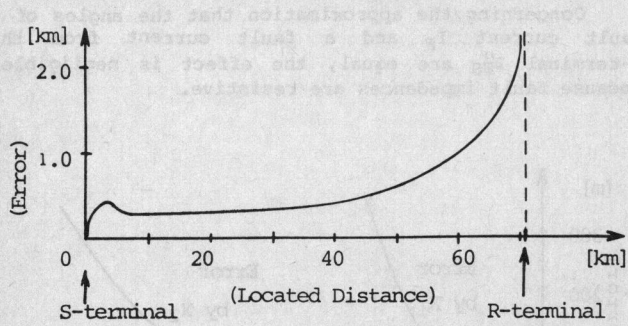


Fig. 4 An example of error caused by an overhead ground wire current

CONFIGURATION

Figure 5 shows the block diagram of this equipment. This equipment has the capacity to calculate, at the same time, for two transmission lines used on the same tower. The voltages of two transmission lines are transformed to a Y quantity for a ground fault and a Δ quantity for a phase-to-phase fault, to minimize calculation error. When a fault occurs on a transmission line, the calculated distance to the fault point and the fault phases are indicated on the display panel, and some of calculated data are printed out by a printer. The data printed out are as follows:

- i) Located distance and fault phase(s).
- ii) Pre-fault and after-fault voltage and current
- iii) The changes of calculated result from the time the fault was detected to the time the result was converged.

All mutual- and self-impedances of each of the lines are input from the setting panel. A picture of the DFL is shown in Fig. 6.

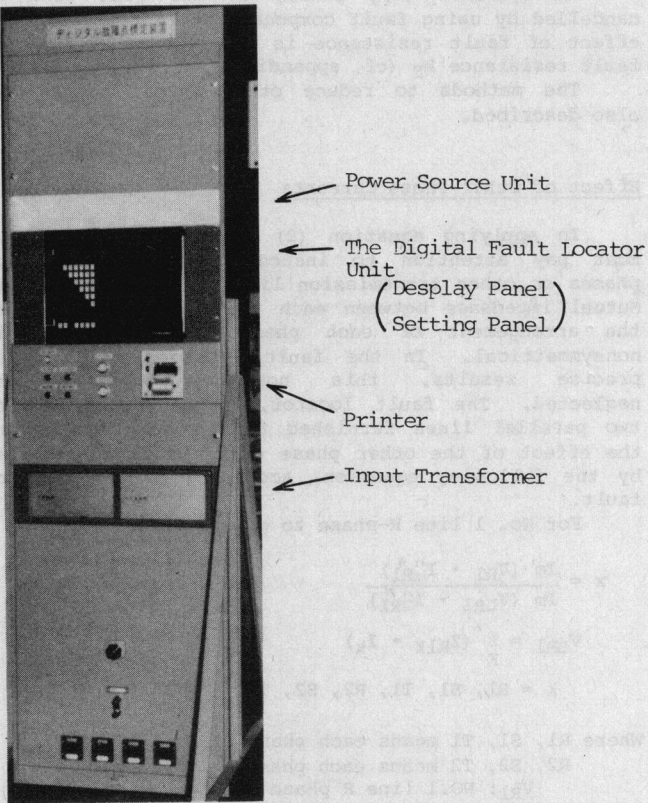


Fig. 6 Front view of the digital fault locator

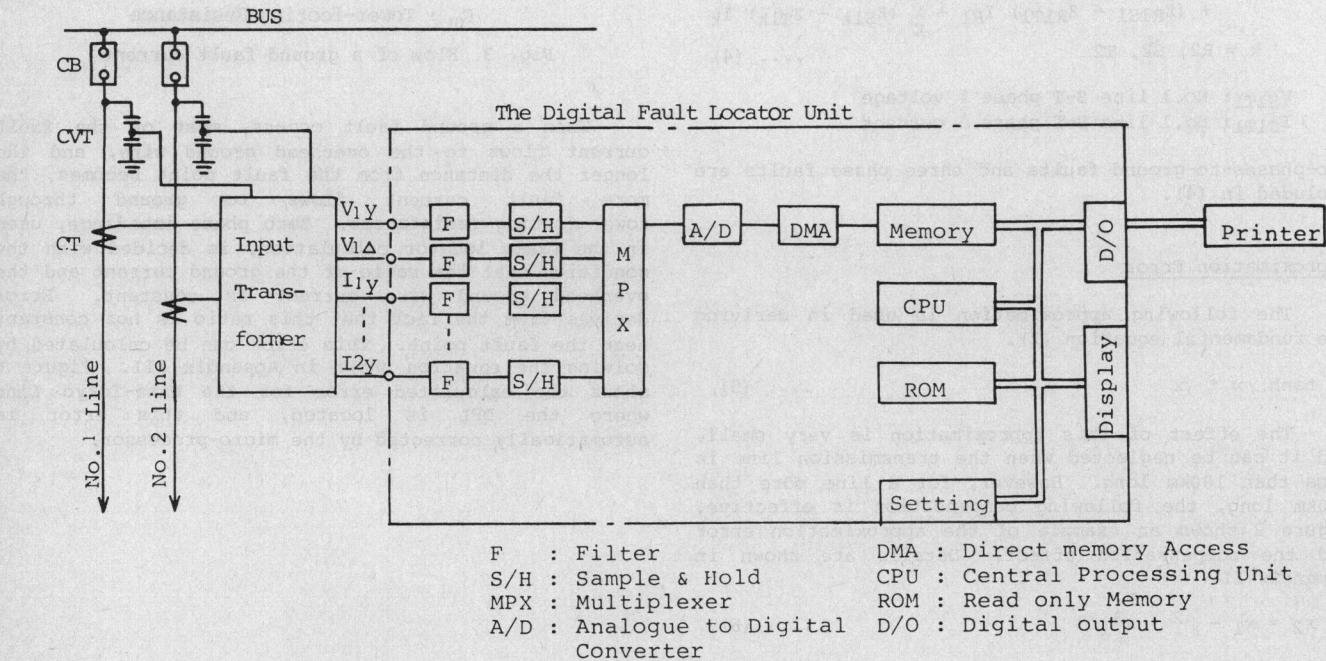


Fig. 5 The block diagram of the digital fault locator

ALGORITHM

Figure 7 shows the flowchart of the digital fault locator (DFL). In the normal state, the CPU calculates the under-voltage relay elements of each phase to detect a fault on the transmission line. The characteristic of this under-voltage relay is compensated by the fault current, as shown in Fig. 8. It can be easily used for a heavy load, long distance transmission line. When a fault occurs on a transmission line, the data of pre-fault current are stored immediately, and the fault phases are selected by the operating mode of the under-voltage relays.

Fault locating calculations are executed only about the faulty phases every 5 ms. The calculation results are recognized as converged when the serially calculated three results coincide within $\pm 100\text{m}$. After the calculation results are converged, the result is displayed and printed out.

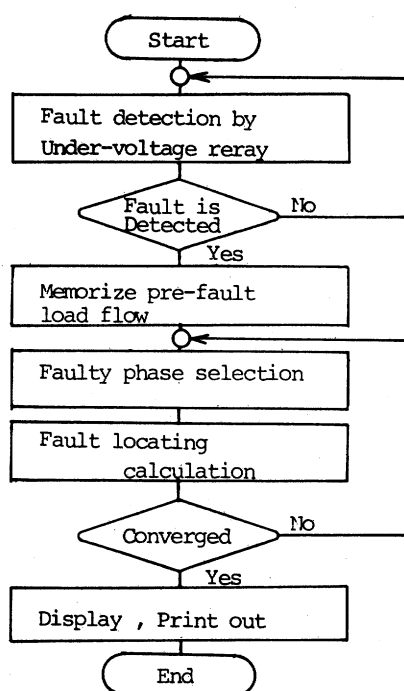


Fig. 7 Flowchart

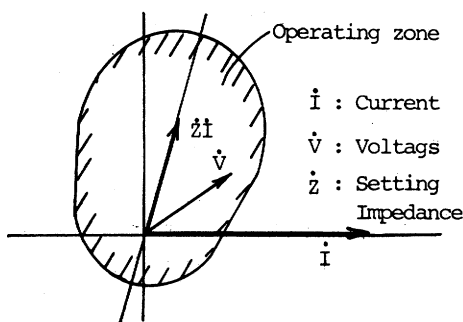


Fig. 8 Characteristic of the under-voltage relay with current compensation

According to the factory test, it takes 50 - 70 ms from the fault occurrence to the result convergence. During this transient period, the calculation routine covers fault phase selecting functions, to deal with changes of fault mode. In case, the fault is cleared before the convergence of calculation, the result that is nearest to the convergence is displayed.

FIELD TEST

Test Results

The field test of the DFL was started in November 1980 at the Naka-Tokyo trunk line in the Shinanogawa Power Station of the Tokyo Electric Power Co. (TEPCO). The field test is now going on and will continue until September 1982. The Naka-Tokyo trunk line is 71.2 km long and consists of two parallel transmission lines.

In the winter of 1981, there was very heavy snow on the Naka-Tokyo trunk line, and we experienced eight faults in January and February. These eight faults were all caused by snow falling on the insulators and breaking down the insulation. The snow was so heavy that the traces of the fault could not be inspected immediately. Inspection was performed later in June and traces of the fault were evident by the marks of arcs made by the fault on each insulator.

Table 1 shows the comparison between the fault locating results and the inspection results. Case 9 fault occurred in July, by lightning.

In six cases, (Case 2, 3, 4, 5, 7, and 9) the error was within 0.9 km and the results were very satisfactory.

The marks of an arc corresponding to Case 1 were not discovered. Case 6 was a multi-fault that occurred at the same time on different points. For this type of multi-fault, correct results cannot be expected. In case 8, the calculation result had not converged before the fault was cleared. It is considered for the reason that the fault resistance, mainly affected by thickly lied snow, was changed during the fault, and fault current fluctuated.

Figure 9 shows the changes of calculation results from the time the faults were detected to the time the results were converged in the representative cases.

Performance for Multi-Faults

This equipment was designed to calculate the distance precisely for a single fault on a transmission line. However, when a multi-fault occurs simultaneously at different places, then the correct result cannot be expected. Case 6 of the field test was a good example. In this case, R-phase was grounded at the 49.0 km point. T-phase was grounded at the 34.2 km point at the same time, and the DFL indicated 46.6 km-RT-2 phase-to-ground fault.

In this case, the fault current was not balanced; 4160A in R phase and 4780A in T phase. By paying attention to this unbalanced current, it is possible to distinguish multi-faults occurring at different points from a single fault. For Case 4, which was a RT-2-phase-to-ground fault, the two phase fault currents were balanced; 4090A in R-phase, 4030A in T-phase. The two different fault points can be roughly calculated by using the ratio of the two different faulty currents. In Case 6, we calculated the two fault points from the ratio of each phase current; 49.3 km for R-phase, and 35.7 km for T-phase. It was very close to the real fault point. (Cf. Table 1)

Table 1 The DFL locating results and the inspection results

Case	The DFL Locating Results (A)						The inspection Results (B)			Error (A-B) Km	Error (%)
	Date	Time	Line No.	Kind of Fault	Locating Result	Tower No.	Distance	Tower No.	Faulty Phase		
1	'81-Jan-18	11 ⁰ 57'	1L	S-1 ϕ G	25.7Km	109	Not found			*	*
2	'81-Jan-19	1 ⁰ 58'	2L	R-1 ϕ G	41.8Km	71	41.3Km	72	2L-R	0.5Km	1.2%
3	'81-Jan-20	11 ⁰ 34'	1L	S-1 ϕ G	54.7Km	38	55.1Km	37	1L-S	-0.4Km	0.7%
4	'81-Jan-21	9 ⁰ 30'	2L	RT-2 ϕ G	32.8Km	91	32.6Km	92	2L-RT	0.2Km	0.6%
5	'81-Jan-21	19 ⁰ 42'	2L	T-1 ϕ G	44.9Km	61	45.3Km	60	2L-T	-0.4Km	0.9%
6	'81-Jan-22	16 ⁰ 06'	2L	RT-2 ϕ G	(46.6Km)	(56)	49.0Km 34.2Km	51 88	2L-R 2L-T	*	*
7	'81-Jan-31	13 ⁰ 02'	1L	T-1 ϕ G	31.0Km	96	30.2Km	98	1L-T	0.8Km	2.6%
8	'81-Feb-3	18 ⁰ 31'	1L	S-1 ϕ G	(40.9Km)	73	38.5Km	77	1L-S	*	*
9	'81-Jul-12	16 ⁰ 35'	2L	T-1 ϕ G	70.7Km	3	69.8Km	5	2L-T	0.9Km	1.3%

*Case 1: Marks of an arc were not discovered.

*Case 6: Multi-Fault at the different points.

*Case 8: Not converged

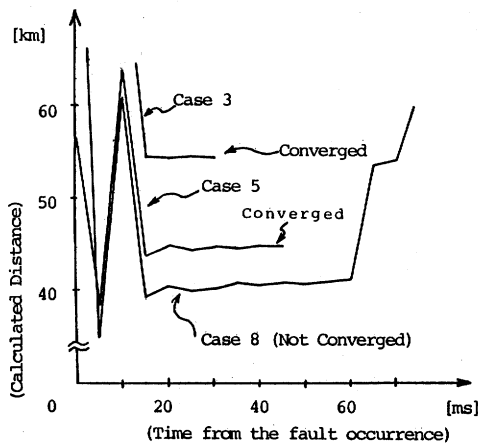


Fig. 9 The changes of calculation results

CONCLUSION

Fault location using only one terminal voltage and current data, which was considered difficult to obtain, was accomplished by using a micro-processor. All types of error factors are analyzed and corrected by using the advantages of software calculations.

The long term field test of the DFL, using these newly developed method, indicated outstandingly accurate locating results. Considering that the DFL can be easily realized by only preparing CT and VT signals of one terminal, we believe that it will be widely used in the near future.

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- [2] T.Takagi, et. al. "A New Algorithm of An Accurate Fault Location for EHV/UHV Transmission Lines: Part II - Laplace Transform Method." IEEE PES Summer Meeting 81 SM 411-8
- [3] Y.Akimoto, et. al. "Micro-processor Based Digital Relays Application in TEPCO." IEEE PES Winter Meeting, 81 WM 119-7, July 1980.

APPENDIX I

A single phase circuit with a fault as shown in Fig. Al.1 (a) can be separated into two equivalent circuits: The pre-fault load flow component shown in Fig. Al.1 (b) and the fault current component shown in Fig. Al.1 (c). From the fault point

$$V_F = I_F R_F = (I_{SF} + I_{RF}) R_F = (I_{SF}'' + I_{RF}'') R_F + V_{dp} \quad \dots (Al.1)$$

V_F and I_{SF}'' are expressed using values which can be observed from the S-terminal.

$$V_F = V_S \cosh \gamma x - Z_S I_S \sinh \gamma x$$

$$I_{SF}'' = V_S' / Z_S \sinh \gamma x - I_S'' \cosh \gamma x \quad \dots (Al.2)$$

where

- x : distance to the fault point
- V_S : voltage of S-terminal
- I_S : current of S-terminal
- V_F : voltage of fault point
- I_F : fault current
- I_{SF} : fault point current from S-terminal
- I_{RF} : fault point current from R-terminal
- V_S' : voltage difference between pre-fault and after-fault voltage.
- I_S'' : current difference between pre-fault and after-fault (fault component current)
- I_{SF}'' : fault current from S-terminal
- I_{RF}'' : fault current from R-terminal
- R_F : fault resistance
- Z_S : surge impedance
- Z : transmission line impedance per unit length
- γ : propagation constant
- Im : imaginary component

Fault current I_F is expressed by the following equation using I_{SF}'' .

$$I_F = I_{SF}'' \cdot \zeta$$

$$\zeta = \zeta \cdot e^{j\theta}, \theta = \arg(I_F / I_{SF}'') \quad \dots (Al.3)$$

(Al.1) (Al.2) (Al.3) gives the following equation.

$$V_S - I_S Z_S \tanh \gamma x + (V_S' / Z_S \tanh \gamma x - I_S'') \zeta \cdot e^{j\theta} = 0 \quad \dots (Al.4)$$

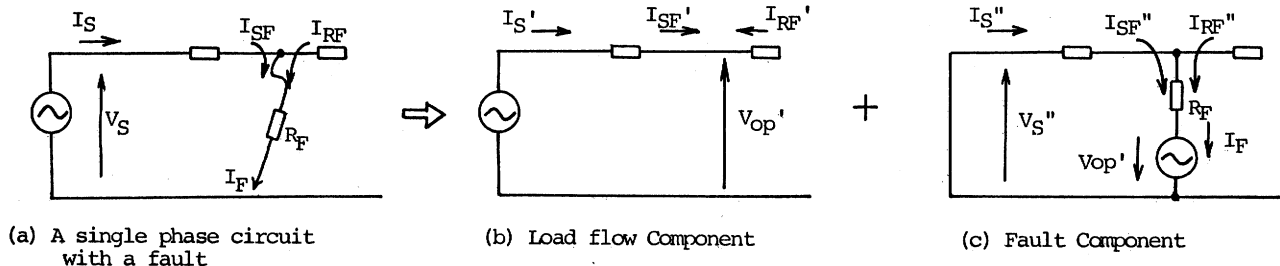


Fig. A1.1 Two equivalent circuits for a fault

From (A1.4), ζ can be eliminated.

$$\text{Im} \{ (V_S - I_S Z_S \tanh \gamma x) (V_S' / Z_S \tanh \gamma x - I_S')^* \epsilon^{-j\theta} \} = 0 \quad \dots (A1.5)$$

* expresses conjugated component

In (A1.5), unknown values are θ and x . When θ is given, then the distance to the fault point x can be obtained. θ is a value which represents a phase angle difference between the fault currents from the S- and R-terminal. It's value is expected to be zero or near zero. In (A1.5), replacing θ by zero, and using the next approximation (A1.6), x can be solved by (A1.7).

$$\tanh \gamma x \approx \gamma x \quad \dots (A1.6)$$

$$V_S' / Z_S \tanh \gamma x \ll I_S'$$

$$x = \frac{\text{Im} (V_S I_S'^*)}{\text{Im} (Z_S I_S'^*)} \quad \dots (A1.7)$$

where $Z = \gamma Z_S$: Impedance per unit length.

APPENDIX II

Define $f(x)$ to be the left term of (A1.5) in Appendix I. x_1 means approximate solution by (2) and x_2 means correct solution by (A1.5). Then, error by approximation Δx is expressed as follows:

$$x_2 = x_1 + \Delta x \quad \dots (A2.1)$$

$$\text{or} \quad \Delta x = -f(x_1) / f'(x_1) \quad \dots (A2.2)$$

Approximation is expressed using small value δ_1 , δ_2 , as follows:

$$\left. \begin{aligned} \tanh \gamma x &= \gamma x + \delta_1 \\ V_S' &= jX_B I_S', X_B / Z_S = \delta_2 \end{aligned} \right\} \quad \dots (A2.3)$$

where X_B : Back impedance of the transmission line

From (A2.3) and (A1.5), $f(x_1)$ and $f'(x_1)$ can be obtained using the next approximation:

$$\delta_1 = \tanh \gamma x - \gamma x = -(\gamma x)^3 / 3 + 2 (\gamma x)^5 / 15 \dots \dots (A2.4)$$

Then the following solution can be obtained from (A2.2).

$$\Delta x = -\frac{\beta^2 x_1^3}{3} + \frac{\beta^3 x_1^5}{5} \dots \dots (A2.5)$$

Neglecting the higher order term, then:

$$x_2 = x_1 - \frac{\beta^2 x_1^3}{3} \quad \dots (A2.6)$$

APPENDIX III

From Fig. A3.1,

$$\left. \begin{aligned} -\frac{\partial V}{\partial (x/S)} &= I \cdot \zeta_\ell - I_{SF}'' \cdot \zeta_m \\ -\frac{\partial I}{\partial (x/S)} &= V / R_T \end{aligned} \right\} \quad \dots (A3.1)$$

where

- ζ_ℓ : Self impedance of ground and overhead ground wire per single span(s)
- ζ_m : Mutual impedance between overhead ground wire and the transmission line.
- R_T : Tower-footing resistance
- r_0 : Mesh grounding resistance in Sub-Station

Because r_0 is negligibly small compared with R_T , next boundary conditions are given:

$$\left. \begin{aligned} V &= 0 \quad \text{at} \quad x = x_0 \\ I_1 &= I \quad \text{at} \quad x = 0 \end{aligned} \right\} \quad \dots (A3.2)$$

by (A3.2), (A3.1) are solved:

$$\begin{aligned} I &= I_{SF}'' \frac{\zeta_m}{\zeta_\ell} + (I_1 - I_{SF}'' \frac{\zeta_m}{\zeta_\ell}) \cdot \\ &\frac{\epsilon^{-\gamma x/S} + \epsilon^{\gamma(x-2x_0)/S}}{1 + \epsilon^{-2\gamma x_0/S}} \end{aligned} \quad \dots (A3.3)$$

The fundamental equation (2) covers only the first term of the right hand of (A3.3), and the second term of (A3.3) becomes the error factor. The effect of this error factor is shown in Fig. 4.

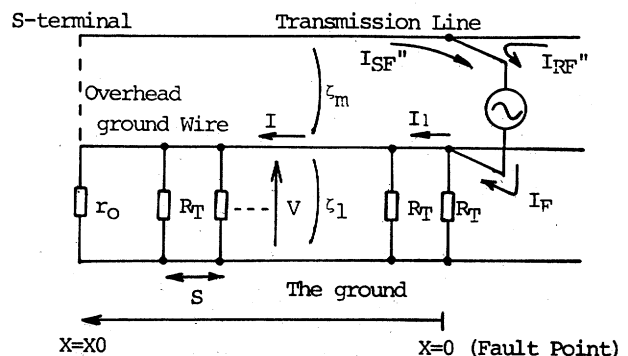


Fig. A3.1 Equivalent circuit for a ground current analysis

Discussion

William Premerlani (GE CR&D, Schenectady, NY): Would the authors please discuss the following question. In Appendix I, equation A1.6, two approximations are made. The first is the short line approximation. The second approximation that is made would appear to be true only if the source impedance behind the relay is not too large. Otherwise, the fault current will be very small and the approximation will not be valid. Would the authors please describe how the range of source impedances over which this approximation is valid can be determined?

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M. Yamaura and T. Matsushima: The authors would like to thank Mr. W. Premerlani for his interest.

As described in the paper, the term $V_s/Z_s \tanh \gamma X$ in (A1.5) is neglected, compared with the term I_s . Considering a physical meaning of the neglected term, the term is equivalent to a charging current of line capacitance, caused by the different voltage V_s between pre- and after-fault voltage at S-terminal. Generally speaking, the longer a transmission line becomes, the more a charging current increases. And, as mentioned by the discussor, the larger the source impedance becomes, the

smaller I_s becomes and the more $V_s/Z_s \tanh \gamma X$ (the charging current term) increases. But, in ordinary transmission lines, however larger the source impedance may be, the charging current is almost negligibly small compared with the fault current. And further, we must notice that the charging current is capacitive, while the fault component current is inductive.

Now, consider an effect of I_s' in the fundamental equation (A1.7). It can be seen easily that an error is absolute value of I_s' does not affect the result, because the denominator and numerator contain I_s' in common. Only the angle error of I_s' caused by the neglected charging current term can affect the result. Considering that the I_s is inductive and a charging current is capacitive, error caused by the charging current term is expected to be extremely small.

In Appendix II, the approximation error is estimated, and the estimation is based on (A1.5). So, the result shown in Fig. 2 contains the effect of the neglected charging current term. Figure 2 shows that (2) can be used for a transmission line under 100km, and the compensation by (6) is effective for 100-300km transmission line. For more than 300km transmission line, more precise compensation will be needed. The error shown in Fig. 2 is mainly affected by a line length rather than a source impedance.

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